

A Big Data Framework for Sleep Health: Predictive Modeling and Behavioral Clustering

Final Project Report

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I. Introduction

Sleep plays an essential role in our overall health, yet many people experience sleep-related issues that impact daily performance and long-term well-being. Our group sought to explore how measurable characteristics such as lifestyle habits, biological attributes, and demographic factors relate to sleep outcomes. To do this, we analyzed a dataset consisting of 374 individuals, which includes features such as sleep duration, sleep quality, physical activity level, heart rate, blood pressure, BMI category, gender, age, occupation, and daily steps. These variables allow us to examine whether certain trends or patterns may help distinguish individuals with healthy sleep behaviors from those experiencing Insomnia or Sleep Apnea.

Before building any models, we conducted several data preprocessing steps to ensure accuracy and consistency. Missing values were handled, categorical variables were standardized, and unnecessary identifiers were removed. We then performed exploratory data analysis (EDA) to understand the underlying characteristics of the dataset. This included visualizing how sleep disorder categories vary by gender and occupation, analyzing correlations among numerical features, and reviewing distribution patterns using boxplots, pair plots, and heatmaps to identify meaningful initial trends.

After better comprehending our dataset, we applied multiple supervised learning algorithms to predict sleep disorder classifications. We evaluated Logistic Regression, Naïve Bayes, Decision Trees, and K-Nearest Neighbors (KNN). Our group used accuracy,

confusion matrices, and ROC curves to measure model performance. After these evaluations, we understood the features that played a key role in classification accuracy and how each model distinguished between Insomnia, Sleep Apnea, and No Disorder.

In addition to classification, we explored unsupervised learning to identify hidden patterns within the data. We applied Principal Component Analysis (PCA) to reduce dimensionality while retaining most of the dataset's variance, and then implemented K-Means clustering to uncover possible groupings among individuals based on shared lifestyle and physiological factors. These clustering results provided an additional layer of insight into relationships between habits, health characteristics, and sleep outcomes.

II. Data Explanation

For this project, we used the Sleep Health and Lifestyle Dataset from Kaggle (<https://www.kaggle.com/datasets/uom190346a/sleep-health-and-lifestyle-dataset>) The dataset included 374 entries with information about a person's lifestyle and health. The dataset records variables such as age, gender, occupation, average sleep duration, stress level, physical activity, and daily step count. It also includes physiological measures like BMI category, blood pressure, and heart rate.

The target column shows whether a person has Insomnia, Sleep Apnea, or No Disorder. In the original dataset, many rows had missing values (NA) in the disorder column. To keep the dataset complete, those missing values were reclassified as "No Disorder."

In general, the dataset provides a mix of behavioral and physiological factors, making it useful for studying how lifestyle and health conditions are connected to sleep outcomes.

III. Data Cleaning

We carefully prepared the dataset in advance, before performing any modeling, by making the information complete, consistent, and suitable for analysis. There were 374 observations in this dataset, consisting of demographic, lifestyle, and physiological attributes. While the structure of these data was generally organized, a number of important adjustments were needed to optimize its quality.

One of the first problems we noticed was the missing values in the Sleep Disorder variable. Quite a few of the entries in this column were marked as missing, and according to the documentation, those without a diagnosed condition should be classified as no disorder. With that in mind, every missing value present in the Sleep Disorder column was recoded to No Disorder. This technique kept the target variable meaningful while retaining all the observations.

We removed the Person ID column next, as it was just an identifier and added no predictive value. Next, we standardized some categorical variables whose categories were not fully consistent. For example, the variable BMI Category included two terms for the same thing, Normal and Normal Weight. To eliminate ambiguity, these were combined into a single category labeled Normal. Yet another variable that needed refinement was Blood Pressure, presented originally as a string. In order to put this information into a useful form, we split the systolic and diastolic values into two separate numeric variables.

These steps of cleaning and preparation transformed the dataset into a complete, consistent, and analytically sound foundation. This process supported the reliability of the subsequent exploratory analysis and modeling procedures, ensuring that the findings from this study were based on well-structured and correctly processed data.

IV. EDA

Once the dataset was cleaned, an exploratory data analysis was performed to understand how the variables behaved with each other and how they relate to sleep disorders. We first looked at the distribution of each numerical feature, such as age, sleep duration, physical activity level, stress level, heart rate, and daily steps. Most variables had reasonable ranges without extreme skew, although heart rate had a few high-value outliers. We also examined categorical variables, including gender, occupation, and BMI category, to see how they were represented across the dataset. Visualizations helped us understand the balance of each class, and we noticed that No Disorder was the most frequent label compared to insomnia and sleep apnea.

Then, we compared features across disorder groups using boxplots, countplots, and pairplots. This step revealed meaningful patterns: individuals with insomnia generally slept fewer hours and scored higher on stress, while sleep apnea appeared more common among subjects classified as overweight or obese. Occupations related to healthcare and more demanding roles also showed a great presence of sleep apnea cases. Pairplots allowed us to view relationships between multiple variables simultaneously, and they visually confirmed the clustering tendencies that later benefited models like KNN.

Finally, we generated a correlation heatmap to quantify how numerical variables interact. Sleep duration showed a positive association with sleep quality, suggesting that longer sleep tends to align with better rest. Stress level correlated positively with heart rate and negatively with daily steps. Altogether, the EDA helped understand where the strongest relations in the data were coming from and what features might contribute most to classification. It also gave us expectations that insomnia might relate more to lifestyle factors, while apnea is connected to BMI and cardiovascular indicators.

V. Logistic Regression

To build a strong and interpretable baseline classifier for predicting sleep disorders, we implemented a multinomial Logistic Regression model using the features in our dataset. Because logistic regression is influenced by feature scaling, we standardized the numerical variables such as age, sleep duration, heart rate, and blood pressure. We also encoded categorical predictors and used a workflow so that training and testing data were treated the same way. This approach allowed the model to learn stable class boundaries while reducing the risk of certain variables dominating due to differences in scale.

After training, we evaluated the logistic regression model on the testing set. The classifier achieved an overall accuracy of 0.904 and a balanced accuracy of 0.911, showing strong performance even when accounting for class imbalance. The confusion matrix indicated that most predictions were correct, with particularly strong performance for the “No Disorder” class, where the model correctly classified 49 out of 55 cases. The classification report further confirmed reliable results: the model obtained an F1-score of 0.821 for Insomnia, 0.942 for No Disorder, and 0.889 for Sleep Apnea. Most errors occurred when the model confused Insomnia with Sleep Apnea or misclassified some No Disorder cases as Insomnia. Overall, these results show that logistic regression provided both high predictive accuracy and clear interpretability, making it a strong model for explaining which factors are associated with each sleep disorder category.

VI. Naïve Bayes

The next method we applied was the Naïve Bayes classifier because it is fast, simple, and tends to perform well when features are reasonably independent. Since the dataset contains a mix of lifestyle and biometric variables, Naïve Bayes gives a way to model class probabilities directly and understand how feature distributions differ between “No Disorder”,

"Insomnia", and "Sleep Apnea". It also works efficiently with relatively small datasets, making it a good model to compare against more complex algorithms.

After training the model, Naïve Bayes achieved about 93% accuracy. It showed strong precision and recall for all three classes, especially Sleep Apnea. The model also provided clear insights into which variables were most important for classification. BMI Category and Blood Pressure stood out as key predictors, which makes sense medically, since higher BMI and hypertension are associated with increased apnea risk. Occupation also contributed meaningfully to class separation, suggesting that work-related stress and lifestyle demands influence sleep patterns.

Overall, Naïve Bayes was a valuable model to include. It gave us strong performance with minimal need to tune the data and helped reinforce the relation between physical health indicators and sleep disorder likelihood.

VII. Decision Tree

To better understand how different lifestyle and health factors might influence sleep disorders, we also explored a Decision Tree model. Unlike more complex algorithms, a decision tree works in a very intuitive way: it repeatedly asks simple yes/no questions about the data, such as whether someone's BMI is high or whether their physical activity level is low, and uses those answers to guide the prediction. This makes the model easy to interpret, which was one of the main reasons we included it in our analysis. We limited the depth of the tree to avoid overfitting, while still allowing it to capture meaningful patterns in the data.

When we trained and tested the model, it performed impressively well. The tree reached an accuracy of 93.33% on the testing set. It correctly predicted most cases of Sleep Apnea, Insomnia, and No Disorder, with only a few mistakes overall. The confusion matrix showed that the model had a particularly easy time identifying individuals with Sleep Apnea,

which aligns with the fact that this condition often has clearer physiological markers. Insomnia was slightly more challenging, likely because its symptoms depend more on subjective experiences and lifestyle patterns that can overlap with the other two groups.

One of the most interesting parts of using a decision tree is seeing which variables the model considers most important. The tree relied most heavily on BMI Category, Physical Activity Level, and Blood Pressure, which mirrors what we often hear in medical research: body weight and lifestyle habits play a huge role in sleep health. BMI alone contributed over half of the model's predictive power. Heart Rate, Age, and Occupation also played smaller roles, while features like Stress Level and Quality of Sleep surprisingly had almost no influence on the tree's final decisions. This doesn't mean those factors aren't important in real life, but rather that within this dataset, other variables provided clearer signals for classification.

When we looked at the ROC curves, the model showed strong separation between classes, especially for the No Disorder category. The AUC values, 0.98 for No Disorder, 0.88 for Insomnia, and 0.82 for Sleep Apnea, demonstrated that the decision tree was able to distinguish groups fairly well, even though it wasn't as precise as KNN or Logistic Regression. Still, its main strength was interpretability. Being able to trace the model's predictions back to simple, logical rules helped us understand not just what the model predicted, but why it made those predictions.

Overall, the decision tree added an important layer to our project. While other models delivered slightly higher performance, the tree allowed us to clearly see how factors like BMI and physical activity influence the likelihood of different sleep disorders. This transparency made it a valuable model for interpreting patterns and explaining the relationship between lifestyle habits and sleep health.

VIII. K-Nearest Neighbors (KNN)

Seeking a more flexible classification approach, we used the K-Nearest Neighbors (KNN) algorithm to predict sleep disorders based on the features in our dataset. Because KNN relies on distance measurements, we standardized the numerical variables such as age, sleep duration, heart rate, and blood pressure so that no feature would dominate the model simply due to scale. We also encoded the categorical features and built a preprocessing pipeline to keep everything consistent during training. In order to choose the best version of the model, we ran a grid search with stratified 10-fold cross-validation, testing different values of k and weighting strategies. Throughout this process, KNN performed best when using 5 neighbors with uniform weighting; this model resulted in an average cross-validation accuracy of about 0.873. This tuning step was critical because it prevented overfitting.

We evaluated the model on the testing data once the best hyperparameters were selected. The tuned KNN classifier performed very well, achieving a test accuracy of 94.67%, which was higher than several of the other supervised models in our study. The confusion matrix showed only a small number of misclassifications, and the classification report reflected strong precision, recall, and F1-scores for each sleep disorder type. For example, the model that our group performed reached an F1-score of approximately 0.897 for “No Disorder,” 0.966 for “Sleep Apnea,” and 0.938 for “Insomnia.” In addition, the ROC curve produced extremely high AUC values above 0.99, meaning the model was very accurate in separating the three sleep disorder classes, yielding an almost perfect model.

IX. K-Means

For the last model, a K-Means Clustering algorithm was used to explore natural variation in our data. Without removing the response, the model was used on all the variables. Before moving forward, the variables were once again standardized and checked for outliers

since K-means is a Euclidean distance-based algorithm. The few outliers that appeared in the data were deemed as plausible in the end, as they could represent meaningful, uncommon occurrences. Additionally, Principal Component Analysis (PCA) was also applied in order to reduce dimensions due to our high number of categorical variables (25). It was opted to select a number of 12 components that managed to retain exactly 92.41% of the total variance from the data, essentially reducing our dimensions or variables by 52%.

By using a plot of Total Intra-Cluster Variation against Number of Clusters (K) and the elbow method, it was concluded that the ‘best’ or optimal K was 3 clusters. Finally, by plotting the first and second components against each other due to that they explain the largest amount of variance among all components, we could see that the clusters did not overlap heavily, which tells us that they are well separated. It is highly important to mention that the separations suggest that clusters are far apart from each other in terms of the features that load into the first component and very close together in terms of the features that load into the second component.

The PCA results show that clustering is driven by cardio-metabolic health (PC1) and sleep–stress balance (PC2). Individuals high on both PC1 and PC2 tend to have elevated BP, higher BMI, and sleep apnea, but still maintain good sleep quality, suggesting older yet stable sleepers with chronic conditions. Those high on PC1 but low on PC2 show similar physiological risks combined with poor sleep and high stress. In contrast, low-PC1/high-PC2 individuals have healthy vitals, good sleep, and low stress, forming a “healthy sleeper” cluster, while low-PC1/low-PC2 individuals are physically healthy but experience poor sleep and higher stress.

X. Conclusion

The analysis carried out in this project shed light on how demographic factors, lifestyle habits, and physiological indicators relate to the presence of sleep disorders. This was enabled by careful preparation of the dataset, exploration of its structure, and application of a wide range of supervised and unsupervised learning methods to analyze sleep health from several complementary angles. Each step of the process gave further insight into the variables involved and their interaction.

The feature ranking for the supervised models shows that BMI category, physical activity level, blood pressure, sleep duration, and heart rate were persistently among the most important features that distinguished subjects with Insomnia, Sleep Apnea, and those with no disorder. While all the classification models performed well, each of them emphasized different aspects of the data a little. KNN showed outstanding predictive performance, Logistic Regression and Naïve Bayes yielded stable and interpretable output, while the Decision Tree is intuitively understandable in explaining the decision patterns.

When the complete analysis was run again and all models combined in this final version of the project, some results changed compared to the earlier presented version by the group. These differences occurred because updated preprocessing steps and refined hyperparameters produced estimates that were more accurate and consistent. Certain accuracy values, ROC curves, and feature rankings shifted as a result. These adjustments reflect a more complete and polished analysis, and they strengthened the reliability of the conclusions.

The unsupervised learning methods added further insight. PCA helped reduce the dimensionality while retaining most of the variance in the dataset, and K-Means clustering showed natural groupings among individuals with similar lifestyles and health characteristics.

These clusters aligned with several of the trends observed in the supervised stage, reinforcing the overall patterns found in the data.

The overall findings from the project indicate that sleep disorders stem from biological and behavioral elements, both of which can be effectively identified through machine learning methods. This study once again emphasizes that lifestyle practices and physiological health indicators need to be closely monitored, as they can provide early warning signals associated with sleep-related problems. Integration of multiple approaches in this project enables comprehensive knowledge regarding factors associated with sleep health and encourages potential research on the subject matter.